

Spectral Stability of Pseudoinverse-Based Extreme Learning Machine

Nguyen Bich Van^{1,*}, Khong Ngoc Anh²

^{1,2}: Institute for Artificial Intelligence, VNU University of Engineering and Technology, Vietnam

¹: nbvan@vnu.edu.vn, ²: 22022549@vnu.edu.vn

Abstract

Extreme Learning Machine (ELM) computes output weights analytically using the Moore–Penrose pseudoinverse. Although this leads to fast training, its numerical stability depends strongly on the conditioning of the hidden-layer matrix. This paper studies pseudoinverse-based ELM from a spectral perspective. We show that the smallest singular value governs perturbation amplification in the output weights, while the condition number provides a quantitative measure of hidden-layer instability. We compare SVD-based pseudoinverse computation with iterative hyperpower methods and discuss width-dependent conditioning through a random feature interpretation. Experiments on synthetic matrices and ELM benchmarks show that SVD-based methods remain the most reliable under ill-conditioning, while iterative methods are more sensitive to spectral properties. The results suggest that ELM stability is fundamentally governed by the singular-value structure of the hidden-layer matrix.

Keywords: Extreme Learning Machine, Moore–Penrose inverse, numerical stability, singular values, conditioning

1. Introduction

Extreme Learning Machine (ELM), originally introduced by Huang et al. [3], is a single-hidden-layer feedforward neural network in which hidden-layer parameters are randomly assigned and output weights are computed analytically. Given a training set $\{(x_j, t_j)\}_{j=1}^N$, the hidden-layer matrix is

$$H_{ji} = g(w_i^T x_j + b_i), \quad (1)$$

where w_i , b_i , and g denote the hidden weights, biases, and activation function, respectively. The output weights are obtained from

$$H\beta = T, \quad \beta = H^+T, \quad (2)$$

where H^+ is the Moore–Penrose pseudoinverse.

This closed-form training is computationally attractive, but it is also sensitive to the spectral properties of H . If the hidden-layer matrix is ill-conditioned, small singular values may amplify perturbations in the targets, numerical round-off, or solver approximation errors. Thus, the stability of ELM is not only a learning problem but also a numerical linear algebra problem.

Related work on ELM has emphasized fast analytical training and classification performance [3]. The theory of generalized inverses and least-squares solutions is classical [1], while SVD-based conditioning and stability analysis are standard topics in numerical linear algebra [2, 6]. Iterative inverse and pseudoinverse methods, including Newton–Schulz and hyperpower iterations, may be computationally attractive but can be sensitive to initialization and ill-conditioning. In parallel, random matrix theory provides useful estimates for the singular values of random feature matrices [4, 5].

The contribution of this paper is a compact spectral interpretation of pseudoinverse-based ELM stability. We connect:

hidden-layer spectrum $\rightarrow$ conditioning $\rightarrow$
pseudoinverse sensitivity $\rightarrow$ learning stability.

We also summarize experiments showing how singular values, condition numbers, and solver behavior are related in synthetic matrices and ELM benchmarks.

2. Spectral Stability Analysis

Let $H \in \mathbb{R}^{N \times L}$ be the hidden-layer matrix of an ELM and suppose its compact singular value decomposition is

$$H = U_r \Sigma_r V_r^T, \quad (3)$$

*Corresponding author

where $r = \text{rank}(H)$, $U_r \in \mathbb{R}^{N \times r}$, $V_r \in \mathbb{R}^{L \times r}$ have orthonormal columns, $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$ and $\sigma_{\max}(H) \geq \sigma_1 \geq \dots \geq \sigma_r = \sigma_{\min}(H) > 0$ are positive singular values of H . The Moore–Penrose pseudoinverse is

$$H^+ = V_r \Sigma_r^{-1} U_r^T. \quad (4)$$

Hence,

$$\|H^+\|_2 = \frac{1}{\sigma_{\min}(H)}. \quad (5)$$

Equation (5) shows that the smallest singular value is the key spectral quantity controlling pseudoinverse amplification.

The condition number of H is

$$\kappa(H) = \frac{\sigma_{\max}(H)}{\sigma_{\min}(H)}. \quad (6)$$

It measures how unevenly H stretches different directions. If $\kappa(H)$ is large, at least one direction is nearly collapsed by H , making the least-squares solution sensitive to perturbations.

Consider a perturbation $T \mapsto T + \Delta T$. The corresponding perturbation in the ELM output weights is

$$\Delta\beta = H^+ \Delta T. \quad (7)$$

Therefore,

$$\|\Delta\beta\|_2 \leq \|H^+\|_2 \|\Delta T\|_2 = \frac{\|\Delta T\|_2}{\sigma_{\min}(H)}. \quad (8)$$

Thus, small $\sigma_{\min}(H)$ or large $\kappa(H)$ implies stronger perturbation amplification. This provides the main theoretical explanation for instability in pseudoinverse-based ELM.

3. Solvers and Width-Dependent Conditioning

SVD-based pseudoinverse computation directly uses the singular values of H and is typically backward stable because it relies on orthogonal transformations [2, 6] which preserve Euclidean norm. It also allows small singular directions to be detected and truncated if necessary.

Iterative methods approximate the pseudoinverse through repeated matrix updates. A representative example is the Newton–Schulz iteration

$$X_{k+1} = X_k(2I - HX_k). \quad (9)$$

Such methods can be efficient in favorable settings, but their convergence depends on the initialization, scaling, stopping criteria, and the spectrum of H . When H is ill-conditioned,

convergence may slow down, stagnate, or become sensitive to finite-precision errors.

The hidden-layer matrix in ELM may be viewed as a random feature matrix because

$$H_{ji} = g(w_i^T x_j + b_i) \quad (10)$$

is generated from random hidden parameters such as hidden weights w_i and bias b_i . If the normalized entries of H are independent, centered, sub-Gaussian random variables with unit variance, following a standard theorem in random matrix theory [4, 5] we have

$$\sigma_{\min}(H) \geq c(\sqrt{N} - \sqrt{L}) \quad (11)$$

with probability at least $1 - 2\exp(-C(N - L))$ for some constants $c, C > 0$. Consequently, $\|H^+\|_2 \leq \frac{1}{c(\sqrt{N} - \sqrt{L})}$.

Although practical ELM matrices are not fully independent random matrices, this estimate motivates a width-dependent interpretation: larger sample size tends to improve stability, while excessively large hidden width may deteriorate conditioning.

4. Experimental Results

We conduct two groups of experiments. The first group uses synthetic matrices with prescribed singular-value spectra to compare SVD-based and iterative pseudoinverse solvers. The second group evaluates pseudoinverse-based ELM on benchmark classification datasets, including MNIST, Fashion-MNIST, and ISOLET.

The main evaluation metrics are least-squares residual, Penrose residuals, runtime, convergence success, classification accuracy, the smallest singular value $\sigma_{\min}(H)$, and the condition number $\kappa(H)$.

Table 1 summarizes the convergence behavior of iterative pseudoinverse methods under different spectral conditions. The results show that iterative methods are reliable for well-conditioned and moderately ill-conditioned spectra, but fail completely for severely ill-conditioned spectra. In particular, all iterative runs succeeded in the well-conditioned case, whereas no iterative method succeeded in the severely ill-conditioned case. This confirms that convergence reliability is strongly controlled by the singular-value structure of the matrix.

The detailed residual summaries further support this observation. In the well-conditioned case, all methods achieved

Table 1. Convergence behavior of iterative pseudoinverse methods under different spectral conditions. 10^{-10} and 10^{-7} are thresholds.

Spectrum	10^{-10}	10^{-7}
Well-conditioned	36/36	36/36
Moderately ill-conditioned	36/36	36/36
Severely ill-conditioned	0/36	0/36
Rank deficient	10/36	36/36
Clustered small	24/36	36/36

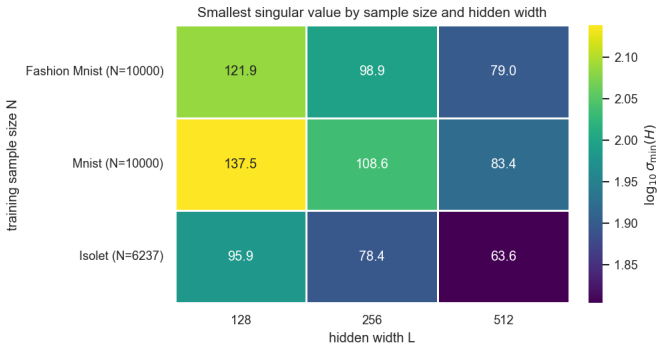


Figure 1. Mean smallest singular value of the ELM hidden-layer matrix for different training sample sizes N and hidden widths L . Larger L generally reduces $\sigma_{\min}(H)$, indicating poorer conditioning and stronger pseudoinverse sensitivity.

successful runs with residuals close to machine precision. For example, Newton–Schulz achieved a median least-squares residual of 1.21×10^{-15} under $\varepsilon_{\text{conv}} = 10^{-10}$. In contrast, under the severely ill-conditioned spectrum, all iterative methods failed, while SVD-based methods retained successful runs.

Figure 1 reports the mean smallest singular value of the ELM hidden-layer matrix under different sample sizes and hidden widths. The results show a consistent decrease of $\sigma_{\min}(H)$ as the hidden width L increases. This behavior supports the width-dependent spectral interpretation: increasing L may improve representational capacity, but it can also worsen conditioning and increase the sensitivity of the pseudoinverse solution.

5. Conclusion and Future Work

This paper studied pseudoinverse-based ELM from a spectral stability perspective. The analysis showed that the smallest singular value determines perturbation amplifica-

tion in the output weights, while the condition number provides a compact measure of hidden-layer ill-conditioning. The comparison between SVD-based and iterative methods indicates that SVD remains the most reliable solver under severe ill-conditioning, whereas hyperpower methods are more sensitive to spectral properties and initialization.

The width-dependent interpretation further suggests that hidden-layer stability depends on the balance between sample size and hidden width. Future work includes regularized ELM, randomized SVD, improved stopping rules for iterative solvers, and larger-scale GPU implementations.

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